# Benjamin Haake | Curriculum Vitae

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A curiosity-driven collaborator, problem-solver, and perpetual learner.

## Education

**PhD in Mathematics** — **Edinburgh**
*University of Edinburgh, supervised by Tudor Dimofte* — *2022–2026*
PhD thesis: Generalised symmetries, orbifold data, and gauging (doi:10.7488/era/7865), and two preprints: arXiv:2506.08178 and arXiv:2605.16942.

**Bachelor and Master of Science in Physics** — **Heidelberg**
*Heidelberg University, BSc 2018, MSc 2021* — *2015–2021*

## Experience in Machine Learning and AI Safety

**Iliad Intensive** — **London**
*LISA* — *7 Sep – 2 Oct 2026*
One-month full-time course on mathematical and conceptual foundations of AI safety.

**Workshop: Maths, Physics, and AI Alignment** — **Edinburgh**
*AGQ Centre for Doctoral Training* — *11–12 May 2026*
Talks and discussions, with speakers from UK AISI, Timaeus, and Stormglass AI, among others.

**Course: Advanced Research Computing** — **Edinburgh**
*AGQ Centre for Doctoral Training* — *Summer Term 2025*
Various applications of machine learning (in PyTorch): particle physics, topological data analysis, geometric structures in biology.

**Course: Deep Learning** — **Bad Hersfeld**
*Summer Academy of the CdE e.V.* — *5–13 Aug 2017*
Basic mathematical foundation of neural networks (e.g. gradient descent) and practical implementation of simple neural networks for image recognition using convolution; in TensorFlow.

## Talks

**Gauging Higher-Group Symmetries** — **Glasgow**
*AGQ Centre for Doctoral Training, AGQ Student Conference* — *4 Nov 2025*

**Gauging 2-group Symmetries in 3d via Orbifold Data** — **Online**
*Beijing Institute of Mathematical Sciences and Applications,* — *20 May 2025*
TQFT and Higher Symmetries Seminar; abstract and video recording available online

**Higher-form Symmetries and How to Gauge Them** — **Vienna**
*University of Vienna, Mathematical Physics Seminar* — *8 Apr 2025*

Various talks in group seminars at the University of Edinburgh and Heidelberg University.

## Further Education

**Conferences and Schools**:
Attendance of 14 conferences and summer schools throughout the 4-year PhD, in Cambridge, Paris, Hamburg, Glasgow, Edinburgh, Glenmore, Bonn, Regensburg, Amiens, Les Diablerets.

**Member of the CdE e.V.** — **Oberwesel, Bad Hersfeld, Windischleuba**
*Week-long "Academies"* — *Since 2015*
Attended university-level courses on quantum algorithms, Fermi problems, English linguistics, inclusive teaching, international law, Judith Butler, designing board games, and more.

## Teaching Experience

**Course Instructor: Algebra and Category Theory** — **Windischleuba**
*Winter Academy of the CdE e.V., about 40 hours* — *28 Dec 2025 – 5 Jan 2026*

**Course Instructor: Differential Geometry** — **Oberwesel**
*Winter Academy of the CdE e.V., about 40 hours* — *28 Dec 2018 – 5 Jan 2019*
**Tutor for undergraduate mathematics and physics courses** — **Heidelberg, Edinburgh**
*Heidelberg University, University of Edinburgh* — *2018–2020, since 2023*

## Additional Skills

### Languages

| | |
|---|---|
| German | First Language |
| English | Fluent (TOEFL iBT score of 116/120) |

### Programming

Python
PyTorch
TensorFlow
Mathematica
LaTeX

Detailed CV available at benninan.github.io.